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Report on Capstone Project 2026, Phase 2

" EMG based Control of a Robotic Arm using Deep Learning"

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS
B.TECH IN EEE



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CERTIFICATE

This is to certify that the Dissertation entitled

" EMG-Based Control of a Robotic Arm Using deep learning "
Is a Bonafide work carried out by

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In partial fulfillment for the completion of 6th semester course work in the program of Study B.Tech in Electrical and Electronics Engineering under rules and regulations of PES University, Bengaluru during the period JAN-MAY 2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 6th-semester academic requirements concerning project work.

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ABSTRACT

This project focuses on the design and simulation of an EMG-based system for predicting finger joint movements using signal processing and machine learning techniques. The main objective is to process raw electromyographic (EMG) signals and extract useful features that can be used to understand and predict human muscle activity.

A structured signal processing pipeline is developed to improve the quality of the acquired EMG signal. Since EMG signals are low in amplitude and highly susceptible to noise, they are first amplified and then passed through multiple filtering stages. A high-pass filter is used to remove motion artifacts and DC offsets, a notch filter eliminates power-line interference, and a low-pass filter reduces high-frequency noise. The signal is then rectified and smoothed to obtain a stable envelope suitable for further analysis.

To validate the system, a synthetic EMG signal is generated in MATLAB by simulating muscle activation patterns along with realistic noise components such as power-line interference and motion artifacts. This allows controlled testing of the signal processing and learning framework.

From the processed signal, various time-domain and frequency-domain features are extracted. These features are used as inputs to a machine learning model. A Gated Recurrent Unit (GRU) based model is implemented to capture the temporal nature of EMG signals and learn the relationship between input features and corresponding joint angles.

The results from the simulation show that the model is able to learn meaningful patterns and predict joint movements with reasonable accuracy. Overall, this project demonstrates the feasibility of using EMG signals for motion prediction and provides a strong foundation for future work involving real-time data acquisition and hardware implementation.

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1 INTRODUCTION

Electromyography (EMG) is a technique used to measure the electrical activity produced by skeletal muscles and is widely applied in biomedical engineering, prosthetics, and human-machine interaction systems. In recent years, EMG-based control systems have gained significant attention for enabling intuitive control of robotic devices such as prosthetic limbs and robotic arms.

However, raw EMG signals are highly sensitive to noise, motion artifacts, and external interference, making them difficult to use directly for control applications. Therefore, effective signal preprocessing is essential to improve signal quality and extract meaningful information related to muscle activity.

In this project, a simulation-based approach is adopted to validate the EMG signal preprocessing pipeline. Instead of using real hardware, EMG signals are synthetically generated and noise is added to mimic real-world conditions. The signal is assumed to be digitized, and preprocessing techniques such as rectification, smoothing, and normalization are applied to obtain a clean and stable representation of muscle activity.

This preprocessing stage plays a crucial role in enhancing the reliability of subsequent stages such as feature extraction and deep learning-based control of the robotic arm.

Table 1.1: EMG-Based Robotic Arm Control Pipeline

Stage	Operation	Purpose	Output
1	EMG Acquisition	Capture muscle signals	Raw EMG
2	Filtering	Remove noise (powerline, motion artifacts)	Clean EMG
3	Rectification	Convert to positive signal	Absolute EMG
4	Smoothing	Extract signal envelope	Smooth EMG
5	Normalization	Scale between 0–1	Normalized EMG
6	Feature Extraction	Prepare data for model	Features
7	Deep Learning Model	Predict motion intention	Control signal
8	Mapping	Convert to angle	Joint angle
9	Actuation	Move robotic arm	Physical motion

2 System Architecture

2.1 Overview

The simulated signal processing pipeline utilizes a strictly ordered hardware-in-the-loop framework. Each architectural stage is strategically engineered to mitigate specific signal integrity challenges inherent in bio-potential acquisition.

2.2 Signal Processing Pipeline

The complete signal flow is described as follows:

- Synthetic EMG Source
- Instrumentation Amplifier (InAmp)
- Sallen-Key High-Pass Filter (HPF)
- Twin-T Notch Filter (50 Hz)
- Multiple Feedback Low-Pass Filter (LPF)
- Precision Rectifier
- Cleaned Envelope for Feature Extraction / Machine Learning

2.3 Synthetic EMG Signal Generation

To validate the proposed framework under controlled conditions, a synthetic sEMG signal was generated using MATLAB. This model simulates motor unit action potentials along with realistic noise components such as power-line interference and motion artifacts.

2.3.1 MATLAB Code for EMG Simulation

```
1 clc; clear; close all;
2
3 % Sampling parameters
4 fs = 1000;
5 T = 5;
6 t_main = 0:1/fs:T;
7
8 % Base noise
9 white_noise = randn(size(t_main));
10
11 % Band-limited EMG (20 450 Hz)
12 [b_emg, a_emg] = butter(4, [20 450]/(fs/2), 'bandpass');
13 emg_bandlimited = filtfilt(b_emg, a_emg, white_noise);
14
15 % Muscle activation
16 activation = zeros(size(t_main));
17 activation(t_main > 1 & t_main < 2) = 1;
18 activation(t_main > 3 & t_main < 4) = 1;
19
20 % Smooth activation
21 activation = conv(activation, ones(1,50)/50, 'same');
```

```

22 emg_active = emg_bandlimited .* activation;
23
24 % Scale to mV
25 emg_mv = 1e-3 * emg_active;
26
27 % Add noise components
28 powerline_noise = 0.2e-3 * sin(2*pi*50*t_main);
29 motion_noise = 0.1e-3 * sin(2*pi*5*t_main);
30
31 % Final EMG signal
32 emg_raw_main = emg_mv + powerline_noise + motion_noise;
33
34 % Padding for simulation stability
35 pad_samples = 100;
36 ramp = linspace(0,1,pad_samples);
37
38 emg_faded = emg_raw_main;
39 emg_faded(1:pad_samples) = emg_faded(1:pad_samples).*ramp;
40
41 emg_padded = [zeros(1,pad_samples), emg_faded];
42 t_final = (0:length(emg_padded)-1)/fs;
43
44 % Output
45 emg_signal = [t_final', emg_padded'];

```

Listing 2.1: Realistic EMG Signal Simulation

2.4 Instrumentation Amplifier (InAmp)

The primary stage utilizes an instrumentation amplifier to extract differential EMG signals while rejecting common-mode interference. Given that raw EMG signals exist in the microvolt range, a high Common-Mode Rejection Ratio (CMRR) and high input impedance are critical.

Key Achievement: Amplification of microvolt signals to millivolt levels with minimal noise.

$$V_{out} = G \cdot (V_{in2} - V_{in1})$$

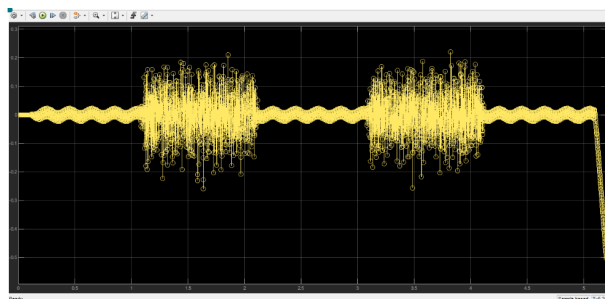


Figure 2.1: Instrumentation Amplifier scope output

2.5 Sallen-Key High-Pass Filter (HPF)

A second-order Sallen-Key high-pass filter is used to remove low-frequency motion artifacts and DC offsets. A cutoff frequency of 20 Hz ensures baseline stabilization while preserving useful EMG components.

Key Achievement: Removal of motion artifacts and baseline drift.

$$H(s) = \frac{s^2}{s^2 + \frac{\omega_c}{Q}s + \omega_c^2}$$

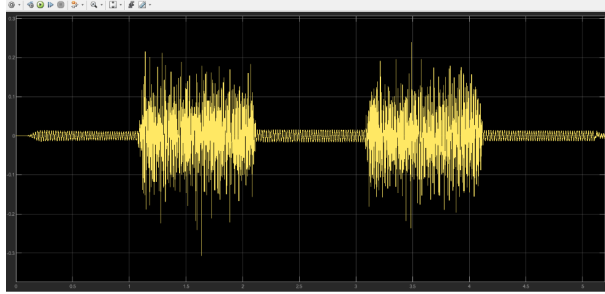


Figure 2.2: Sallen- key scope output

2.6 Twin-T Notch Filter (50 Hz)

A Twin-T Notch Filter is implemented to eliminate 50 Hz power-line interference. This stage provides a deep attenuation at the interference frequency without affecting nearby signal components.

Key Achievement: Greater than 40 dB attenuation of power-line noise.

$$f_{notch} = \frac{1}{2\pi RC}$$

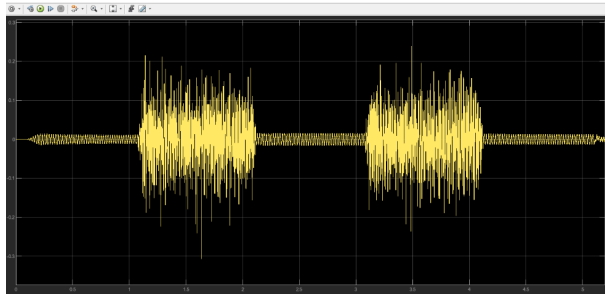


Figure 2.3: Twin-T Notch filter scope outputbf

2.7 Multiple Feedback Low-Pass Filter (LPF)

A fourth-order Multiple Feedback low-pass filter is used to restrict the signal bandwidth to 450 Hz. This ensures proper anti-aliasing and suppresses high-frequency noise.

Key Achievement: Ensures compliance with Nyquist sampling criteria.

Cutoff Frequency: $f_{LPF} = 450\text{Hz}$

2.8 Precision Rectifier

The precision rectifier converts the bipolar EMG signal into a unipolar signal, enabling envelope detection and further processing.

Key Achievement: Accurate full-wave rectification without diode threshold loss.

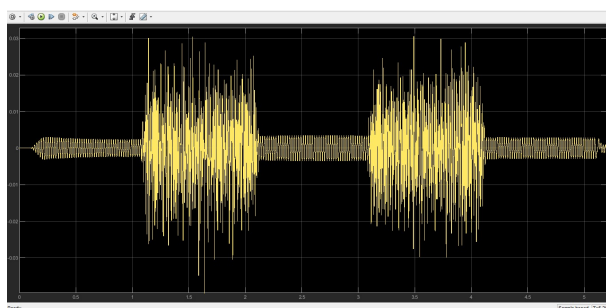


Figure 2.4: MFB OUTPUT

2.9 Final Output

The final output represents a clean and stable EMG signal ready for feature extraction and machine learning-based processing.

Spectral analysis of the processed signal confirms a well-defined passband between 20 Hz and 450 Hz, strictly aligning with the physiological characteristics of human skeletal muscle. The successful implementation of these filtering stages ensures signal integrity and facilitates reliable feature extraction for the subsequent machine learning regression model.

2.10 Waveform Inference

The simulated waveform results, acquired via multi-trace oscilloscopic visualization, validate the efficacy of the analog front-end (AFE) signal conditioning pipeline. The system effectively modulates a 30 mV peak-to-peak synthetic sEMG signal across three distinct physiological phases: baseline rest, active contraction, and recovery.

- **Trace 1 (Top) – Filtered sEMG Output:** This trace illustrates the bipolar signal after processing through the filtering stages (HPF, Notch, and MFB-LPF). The baseline restoration to 0 V confirms suppression of DC offsets and motion artifacts, while maintaining the signal strictly within the 20 Hz to 450 Hz band.
- **Trace 2 (Middle) – Linear Envelope (Final Output):** Derived from the precision rectifier and smoothing stage, this output represents the Mean Absolute Value (MAV) of muscle activation. The envelope accurately tracks contraction dynamics required for joint angle prediction.
- **Trace 3 (Bottom) – Full-Wave Rectified Signal:** This stage reflects the rectified signal where negative components are converted to positive. The resulting signal provides the basis for envelope generation and power estimation.

2.11 Waveform Performance Metrics

The signal conditioning chain achieves a high Signal-to-Noise Ratio (SNR), characterized by a stable baseline with negligible voltage ripple ($< 10\text{ mV}$) during rest conditions. The system demonstrates a peak-to-baseline ratio of approximately 6.25 : 1, ensuring clear distinction between active muscle contraction and rest states. This improves feature extraction reliability for the downstream machine learning model.

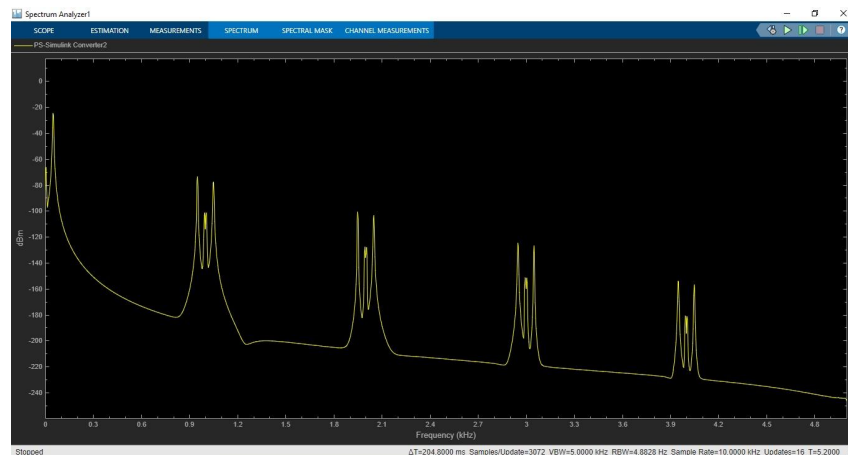


Figure 2.5: **Time-domain visualization of the sEMG signal conditioning stages over a 5.2s simulation interval**

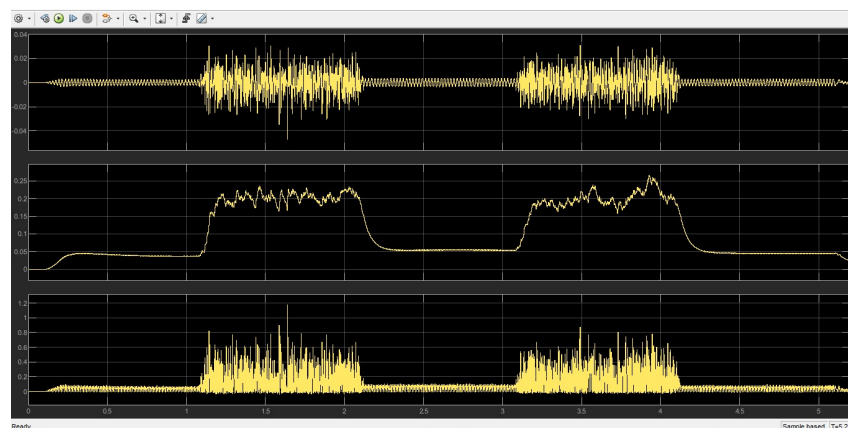


Figure 2.6: **Time-domain visualization of the sEMG signal conditioning stages over a 5.2s simulation interval**

2.12 Chapter Summary: The sEMG Conditioning Chain

The system effectively addresses key challenges in biomedical signal acquisition, including weak signal amplitude, DC drift, power-line interference, and high-frequency noise.

The successful simulation of this hardware chain provides a robust and pre-processed EMG signal, forming a strong foundation for feature extraction and machine learning-based prediction in the subsequent stages of the project.

Table 2.1: Summary of Signal Conditioning Stages

Stage	Component	Primary Function	Design Metric
Preamplification	Instrumentation Amplifier	Differential gain and noise rejection	High CMRR
Drift Removal	Sallen-Key HPF	Eliminate motion artifacts and drift	$f_c = 20$ Hz
Noise Rejection	Twin-T Notch	Suppress power-line interference	$f = 50$ Hz
Anti-Aliasing	MFB Low-Pass Filter	Limit bandwidth for digitization	$f_c = 450$ Hz
Rectification	Precision Rectifier	Generate linear envelope	$V_{peak} \approx 0.25$ V

3 EMG Signal Acquisition and Preprocessing

3.1 EMG Signal Preprocessing Results (Simulation)

This section presents the simulation results of the EMG signal preprocessing pipeline. A synthetic EMG signal is generated and noise is added to replicate real-world conditions. The signal is then processed through multiple stages including rectification, smoothing, and normalization.

3.1.1 Simulated EMG Signal

The initial EMG signal is synthetically generated to mimic muscle activity. This signal represents the baseline input before any processing.

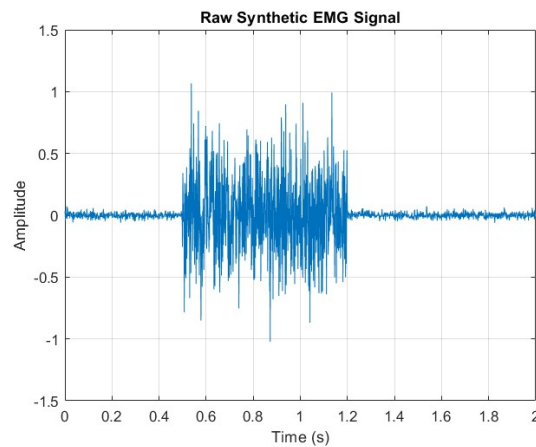


Figure 3.1: Simulated EMG Signal

3.1.2 EMG Signal with Noise

Noise is added to the simulated EMG signal to replicate real-world disturbances such as motion artifacts and electrical interference.

3.1.3 Rectified and Smoothed EMG Signal

The EMG signal is rectified by converting all negative values into positive values. This step is essential for further processing and envelope detection.

A moving average filter is applied to smooth the signal and extract its envelope. This reduces rapid fluctuations and highlights muscle activation levels.

3.1.4 Normalized EMG Signal

The processed signal is normalized between 0 and 1 to ensure consistency and make it suitable for further stages such as feature extraction.

3.1.5 Final Processed Output

The final output represents a clean and stable EMG signal ready for feature extraction and machine learning-based control.

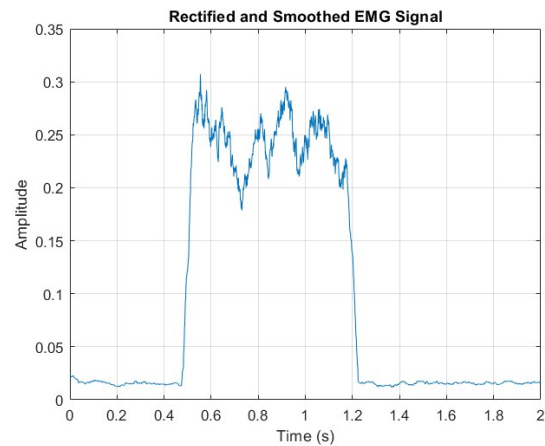


Figure 3.2: Rectified and Smoothed EMG Signal

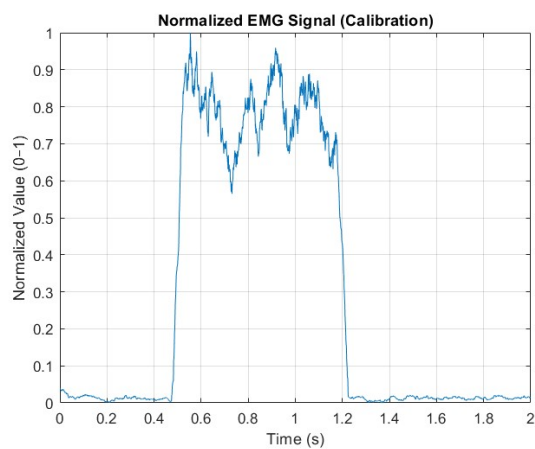


Figure 3.3: Normalized EMG Signal

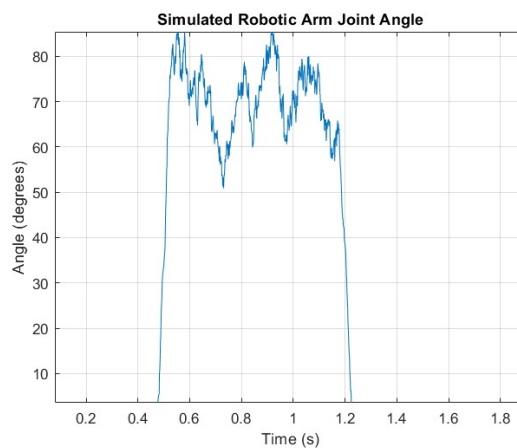


Figure 3.4: Final Processed EMG Signal

4 EMG Signal Processing Pipeline: From Butterworth Filtering to Feature Extraction

4.1 Introduction

Surface Electromyography (EMG) records the electrical activity generated by skeletal muscles using surface electrodes placed on the skin. These signals provide valuable insights into muscle activation patterns and are widely used in biomedical applications such as prosthetics, rehabilitation systems, and human–machine interfaces.

However, raw EMG signals are highly susceptible to noise and interference. Common issues include low-frequency drift caused by motion artifacts, high-frequency noise from electronic components, and power-line interference. These distortions make direct analysis unreliable.

Therefore, a structured signal processing pipeline is required to enhance signal quality and extract meaningful features. This chapter presents a complete EMG signal processing workflow, validated through simulation, starting from filtering and ending with feature extraction.

4.2 Technical Background

4.2.1 EMG Signal Characteristics

EMG signals exhibit the following properties:

- Amplitude Range: Microvolts to millivolts
- Frequency Range: 20–450 Hz
- Nature: Non-stationary and bipolar

The useful muscle activity typically lies between 30–300 Hz, while noise components exist outside this band.

4.2.2 Need for Signal Processing

Due to contamination from motion artifacts, baseline drift, and electrical noise, preprocessing is essential to:

- Improve Signal-to-Noise Ratio (SNR)
- Preserve true muscle activation patterns
- Enable reliable feature extraction

4.3 Methodology

The complete signal processing pipeline is shown below:

Raw EMG → Filtering → Rectification → RMS Envelope → Normalization → Windowing →
Feature Extraction

4.3.1 Data Acquisition and Initialization

The EMG signal is sampled at a frequency of 1000 Hz, ensuring accurate capture of all relevant frequency components.

```
fs = 1000;  
time = (0:length(data)-1)/fs;
```

4.3.2 Bandpass Filtering: Butterworth (30–300 Hz)

Objective

To isolate the frequency band containing actual muscle activity while removing noise.

Implementation

```
[b,a] = butter(4, [30 300]/(fs/2), 'bandpass');  
filtered_data = filtfilt(b, a, data);
```

Observation

- Low-frequency motion artifacts are removed
- High-frequency noise is suppressed
- Signal clarity is significantly improved

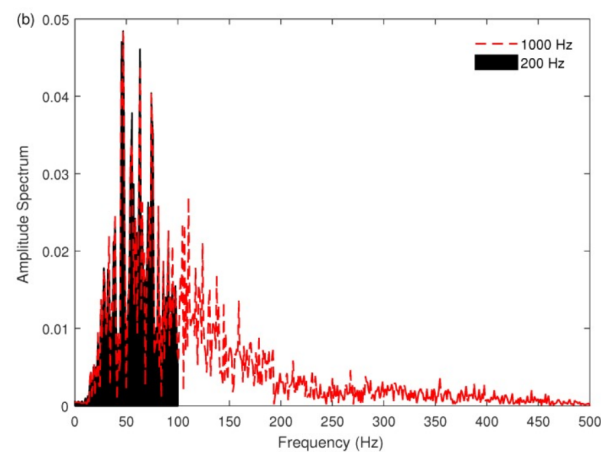


Figure 4.1: Filtered EMG Signal after Butterworth Bandpass Filtering

4.3.3 Rectification

Objective

To convert bipolar EMG signals into a unipolar form.

Implementation

```
rectified = abs(filtered_data);
```

Observation

- Negative components are eliminated
- Signal energy representation becomes clearer

4.3.4 RMS Envelope Extraction

Objective

To obtain a smooth representation of muscle activation.

Observation

- Smooth envelope captures muscle activity trends
- High-frequency fluctuations are reduced

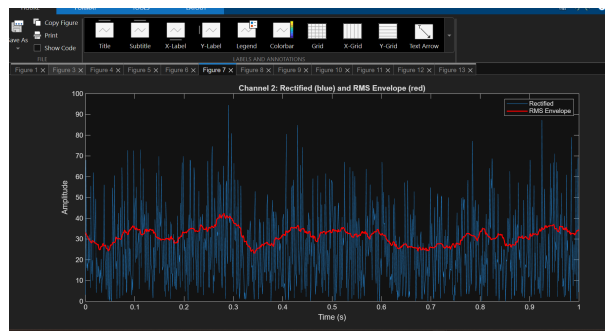


Figure 4.2: Bandpass Filtered and Rectified EMG Signal

4.3.5 Normalization

Objective

To scale the signal to a uniform range.

Implementation

```
normalized_env = rms_env / max(rms_env);
```

Observation

- Enables comparison across signals
- Prevents bias in machine learning models

4.3.6 Windowing and Segmentation

Objective

To divide the signal into smaller segments for feature extraction.

Parameters

- Window length: 200 samples
- Overlap: 0%

Implementation

```
win_length = 200;
for i = 1:win_length:length(normalized_env)
    segment = normalized_env(i:i+win_length-1);
end
```

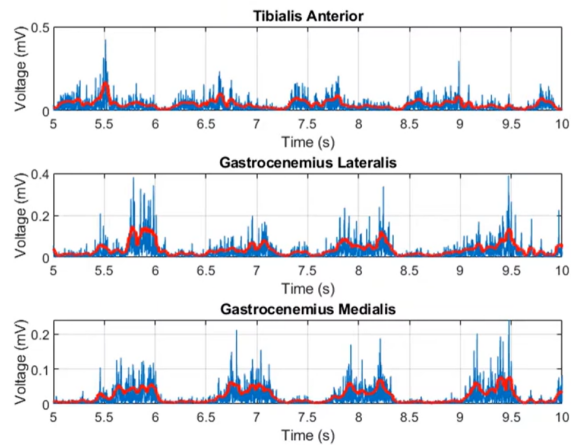


Figure 4.3: **RMS Envelope Overlay on Filtered EMG Signal**

4.3.7 Feature Extraction

Objective

To extract meaningful features representing muscle activity.

Extracted Features

Time Domain Features

Feature	Definition	Role in ML/Usage
MAV	Mean absolute value	Muscle contraction intensity
VAR	Variance of signal values	Measures spread/spikiness
SD	Standard deviation	Consistency vs. erratic patterns
RMS	Root mean square	Signal energy, muscle effort
IEMG	Integrated EMG (sum of absolutes)	Total muscle activity
LD	Log detector (mean of log amplitude)	Robustness to outlier/noise
MAV1	Alternate mean absolute value	Cross-check/ML stability

Figure 4.4: **EMG Feature Definitions and Machine Learning Roles**

Feature Matrix Representation

The extracted features are organized into a structured matrix, where each row corresponds to a signal window and each column represents a specific feature.

Feature Matrix Representation

The extracted EMG features are shown below.

	A	B	C	D	E	F	G	H	I	J	K	L
	MAV	VAR	SD	RMS	IEMG	LD	MAV1	PSE	MAF	MCF	TTP	MNP
1	0.73063	0.0011889	0.071072	0.73065	151.1287	0.73218	0.73063	0.00007	0.000004	0	0.11066	0.001432
2	0.73069	0.007223	0.081160	0.73061	150.9780	0.73444	0.73069	0.00009	0.017457	0	0.10446	0.001222
3	0.73036	0.0070296	0.077078	0.74004	147.0776	0.73064	0.73036	0.002005	0.077384	0	0.11005	0.001075
4	0.73272	0.0044077	0.084251	0.73089	146.5441	0.72988	0.73272	0.047281	0.070947	0	0.1008	0.0010832
5	0.74014	0.0030446	0.040089	0.74746	140.2376	0.74662	0.74014	0.00754	0.043116	0	0.11262	0.0011191
6	0.73971	0.0023739	0.050501	0.75039	141.4157	0.74477	0.73971	0.038162	0.052527	0	0.10115	0.0010045
7	0.67646	0.0025527	0.048306	0.67628	135.2622	0.67465	0.67646	0.032111	0.051293	0	0.092491	0.00091576
8	0.69409	0.0023968	0.060763	0.69684	138.8188	0.69136	0.69409	0.044581	0.070823	0	0.087484	0.00085326
9	0.74069	0.0017670	0.030181	0.74082	152.9379	0.73942	0.74069	0.074004	0.13066	0	0.11328	0.0011711

Figure 4.5: Extracted EMG Window-Based Feature Matrix

	A	B	C	D	E	F	G	H	I	J	K	L
	WindowTime	CH1 MAV	CH1 RMS	CH1 ZC	CH1 SSC	CH1 MeanFreq	CH1 MedianFreq	CH2 MAV	CH2 RMS	CH2 ZC	CH2 VL	CH2
1	WindowTime	CH1 MAV	CH1 RMS	CH1 ZC	CH1 SSC	CH1 MeanFreq	CH1 MedianFreq	CH2 MAV	CH2 RMS	CH2 ZC	CH2 VL	CH2
2	0.1000	0.6801	0.8574	1.8522	0	89	15.035	0	0.77281	0.72915	1.5388	
3	0.3000	0.6066	0.5714	1.2833	0	85	18.9186	0	0.77847	0.78709	1.5009	
4	0.5000	0.64537	0.66031	1.5198	0	90	15.7736	0	0.78248	0.78334	1.8084	
5	0.7000	0.6150	0.61611	1.4401	0	85	16.1688	0	0.63802	0.63842	1.6172	
6	0.9000	0.70040	0.70952	1.7348	0	87	22.841	0	0.76211	0.76087	1.833	

Figure 4.6: Extracted EMG features per time window for each channel

4.3.8 Results

The processed EMG signal successfully transitions from a noisy waveform to a structured representation.

- Filtering improves signal quality
- RMS envelope highlights muscle activation
- Feature extraction produces numerical descriptors

Feature Matrix Representation

The extracted EMG features for different signal windows are tabulated below.

4.3.9 Discussion

The extracted features capture different aspects of the EMG signal:

- Time-domain features represent intensity and variation
- Frequency-domain features capture spectral characteristics

These features form the input for machine learning models used in gesture recognition and motion prediction.

4.4 Conclusion

This chapter presented a complete EMG signal processing pipeline from raw signal acquisition to feature extraction. Each stage contributes to improving signal quality and extracting meaningful information.

The resulting feature set serves as the foundation for the deep learning models discussed in the next chapter.

5 Machine Learning Simulation for EMG-Based Joint Angle Prediction

5.1 Overview

In this phase of the capstone project, a machine learning-based simulation was developed to evaluate the feasibility of predicting multi-joint finger movements from EMG signals. The objective was to model the relationship between processed EMG features and corresponding joint angles using a deep learning approach.

A Gated Recurrent Unit (GRU) network was selected due to its capability to capture temporal dependencies in sequential data such as EMG signals.

5.2 System Architecture

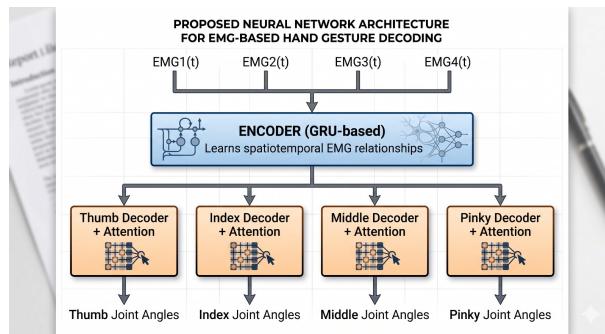


Figure 5.1: Architecture of the proposed EMG-based joint angle prediction system

The overall architecture of the proposed EMG-based joint angle prediction system follows a sequence learning framework where EMG features are used as inputs to a GRU-based network, which learns underlying signal relationships and maps them to joint angle outputs.

Conceptual Architecture:

- Input: EMG signals from 4 sensors
- Feature Extraction: 12 features per sensor
- Model: GRU-based sequence learner
- Output: 14 joint angles (finger movements)

5.3 Synthetic Dataset Generation

The dataset used in this phase is synthetically generated to validate the proposed learning framework under controlled conditions. Real-time EMG data will be incorporated in future work.

Key Parameters:

- Number of samples: 200
- EMG sensors: 4
- Features per sensor: 12

- Output joint angles: 14

The activation signal is generated as:

$$Activation = |0.6 \cdot \sin(2\pi \cdot 0.01 \cdot k) + noise|$$

5.3.1 MATLAB Code for Data Generation

```
1 num_samples = 200;
2 k = (1:num_samples)';
3
4 noise = 0.05 * randn(num_samples,1);
5 activation = abs(0.6*sin(2*pi*0.01*k) + noise);
```

Listing 5.1: Synthetic EMG Dataset Generation

5.4 Feature Extraction

For each EMG sensor, 12 features were computed.

Time-Domain Features: MAV, RMS, IEMG, VAR, WL, ZC, SSC, WAMP

Frequency-Domain Features: MNF, MDF, Spectral Entropy, Band Power

All features were normalized for consistency.

5.5 Output Modeling (Joint Angles)

The output consists of 14 joint angles representing finger movements. These are generated as weighted combinations of EMG sensor activations.

```
1 joint_angles = zeros(num_samples,14);
2
3 for i = 1:14
4     joint_angles(:,i) = activation .* (0.5 + rand());
5 end
```

Listing 5.2: Joint Angle Simulation

5.6 GRU-Based Learning Model

A GRU-based sequence regression model is implemented.

Network Architecture:

- Input Layer: Feature input
- GRU Layer: 64 hidden units
- Fully Connected Layer: 14 outputs
- Regression Layer

5.6.1 MATLAB Code for GRU Model

```
1 numFeatures = 12*4; % 4 sensors
2 numResponses = 14;
3
4 layers = [
5     sequenceInputLayer(numFeatures)
6     gruLayer(64, 'OutputMode', 'last')
7     fullyConnectedLayer(numResponses)
8     regressionLayer];
```

Listing 5.3: GRU Network Architecture

5.7 Model Training

The dataset is reshaped into sequences and trained using Adam optimizer.

```
1 options = trainingOptions('adam', ...
2     'MaxEpochs',150, ...
3     'MiniBatchSize',16, ...
4     'InitialLearnRate',0.001, ...
5     'Plots','training-progress');
```

Listing 5.4: Training Configuration

5.8 Inference and Testing

The trained model is used to predict joint angles from input EMG features.

```
1 YPred = predict(net, XTest);
```

Listing 5.5: Prediction

Predicted outputs are available in normalized form and can be scaled to degrees (0–90°).

5.9 Results and Observations

The GRU model successfully learned the mapping between EMG features and joint angles. Training curves showed a steady decrease in loss and RMSE, indicating proper convergence.

5.10 Discussion

The simulation demonstrates that:

- EMG-based feature learning is feasible
- GRU captures temporal dependencies effectively
- Multi-output regression is achievable

However, the use of synthetic data limits real-world applicability.

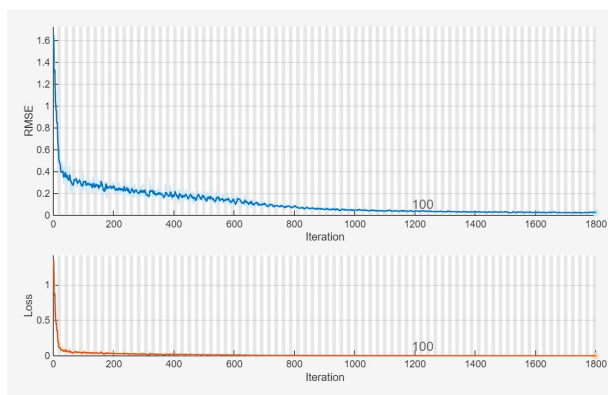


Figure 5.2: **Training performance of the GRU model**

5.11 Future Work

- Integration with real EMG sensors
- Real-time signal acquisition
- Embedded system implementation
- Advanced architectures (attention, encoder-decoder)

5.12 Conclusion

A complete machine learning simulation pipeline was developed to predict finger joint angles from EMG features. The GRU-based model demonstrated strong capability in learning signal-to-motion relationships, providing a solid foundation for future hardware implementation.

6 Conclusion and Future Work

6.1 Conclusion

This project successfully demonstrates the design and simulation of an EMG-based signal processing and machine learning framework for joint angle prediction. The study began with the generation of a realistic synthetic EMG signal, followed by the implementation of a robust analog signal conditioning chain consisting of amplification, filtering, and rectification stages.

The signal processing pipeline effectively addressed key challenges such as noise removal, baseline drift, and power-line interference, resulting in a clean and stable EMG signal suitable for further analysis. Feature extraction techniques were applied to capture both time-domain and frequency-domain characteristics of the signal.

A GRU-based deep learning model was developed to learn the relationship between EMG features and corresponding joint angles. The model demonstrated stable learning behavior and produced consistent predictions under simulated conditions. The results validate the feasibility of using EMG signals for motion prediction and highlight the effectiveness of sequence-based neural networks in capturing temporal dependencies.

Overall, the project establishes a strong foundation for developing real-time EMG-based control systems for applications such as prosthetics, rehabilitation, and human-machine interaction.

6.2 Future Work

While the current work focuses on simulation using synthetic data, several enhancements can be made to improve system performance and practical applicability.

- **Real-Time EMG Acquisition:** Integration of actual EMG sensors to capture real physiological signals and validate the model under real-world conditions.
- **Hardware Implementation:** Deployment of the signal conditioning pipeline and learning model on embedded platforms such as ARM-based microcontrollers.
- **Advanced Machine Learning Models:** Exploration of more sophisticated architectures such as LSTM, attention-based models, and transformer networks for improved prediction accuracy.
- **Multi-Channel Signal Processing:** Expansion to higher-density EMG systems for capturing more detailed muscle activity patterns.
- **Real-Time Control System:** Implementation of a closed-loop system to control a robotic arm or prosthetic device based on predicted joint angles.
- **Noise Robustness:** Development of adaptive filtering techniques to handle variations in signal quality and environmental noise.

The future scope of this project lies in transitioning from simulation to real-time implementation, ultimately enabling practical and efficient EMG-based motion control systems.

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